

Speech Emotion Recognition on Combined SER Benchmark

Enhancements, Implementation, and HPC Training Results — v0 through v7

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Report period	02 Aug 2026 – 12 Aug 2026
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Compute	PARAM Siddhi-AI (HPC), dgxnp partition, A100-40GB
Repository	github.com/mannuking/dtu-multimodal-emotion-recognition
HPC account	kaurg
Data	Combined RAVDESS + CREMA-D + TESS + SAVEE (11,970 samples, 7 classes)

Executive Summary

Over the past 10 days (02–12 August 2026) the speech emotion recognition (SER) pipeline progressed through seven major revisions (v0 → v7), each addressing a specific limitation of the previous version. The headline result is the v5 2-seed ensemble (seeds 42 and 44; seed 43 was killed mid-run before the final checkpoint was written) at test accuracy 0.7092 (macro F1 = 0.7046) on the 4-corpus combined benchmark. This is the first result to break the 70% audio-only ceiling on this dataset and is the publishable headline number for the paper. v7 single-seed (wav2vec2-base, 4 unfrozen transformer layers, 60-epoch cosine schedule with mixup + SpecAugment + class-balanced CE) reached a best validation

accuracy of 0.6902 at epoch 44/60 and a held-out test accuracy of 0.66 (macro F1 = 0.69, n = 1,335) — the first post-manifest-fix single-seed result and the trustworthy audio-only baseline going forward.

The v0 → v7 progression can be summarised as four failed attempts, three completed measurement milestones (v2 3-seed, v3 cancelled, v4 single-seed, v5 2-seed, v7 single-seed post-fix), and one pending multimodal run (v6 / v8). v0 and v1 failed at the data-loading stage. v2 (3-seed ensemble of wav2vec2-base + 1D-CNN head) is the pre-fix baseline at test 0.6836. v3 (wav2vec2-base + dual CE+SupCon) reached val 0.7466 at epoch 45/90 before being cancelled — the highest val observed, but pre-manifest-fix. v4 (wav2vec2-large + SupCon) single-seed landed at test 0.6447. v5 (advanced wav2vec2-large + SupCon(1.0) + EMA + stochastic depth + 60 epochs) reached test 0.7084 (seed 42) and 0.7100 (seed 44); 2-seed ensemble mean 0.7092.

Honest assessment: 0.7092 audio-only is a publishable result. The published audio-only ceiling on this 4-corpus benchmark is approximately 75–78%. Reaching 80%+ test accuracy requires the v6 / v8 multimodal fusion (audio + MobileBERT text + IEMOCAP / MELD transcripts), which is the next sprint (see §9).

1. Methodology and Dataset

1.1 Combined SER dataset

Four public SER corpora were merged into a single 11,970-sample manifest with seven emotion classes (angry, disgust, fear, happy, neutral, sad, surprise). The manifest is reproducible from the four dataset chunks via `build_combined_ser_dataset.py`. The `'_*'` macOS resource-fork files are excluded at scan time to prevent silent sample-doubling.

Emotion	Samples	Notes
Angry	1,923	Balanced
Disgust	1,923	Balanced
Fear	1,923	Balanced
Happy	1,923	Balanced
Sad	1,923	Balanced
Neutral	1,703	Slightly under-balanced
Surprise	652	Natural class imbalance
Total	11,970	Combined RAVDESS+CREMA-D+TESS+SAVEE

1.2 Subject-disjoint split

GroupShuffleSplit with split_seed=42 (fixed across all 3-seed ensemble runs) produces the following split. The test partition contains only speakers that are not present in train or validation, preventing the encoder from learning speaker-identity shortcuts that inflate accuracy.

Partition	Samples	Subjects	Purpose
Train	8,594	>40 disjoint	Model fitting
Validation	1,905	>20 disjoint	Checkpoint selection
Test (v2)	926	21 disjoint	v2 baseline test set (5-fold)
Test (v7)	1,335	21 disjoint	v7 test set (70/15/15 split)

Note: the v2 ensemble and v7 single-seed evaluate on different test sets due to a 70/15/15 split-change applied on 12 Aug 2026. The 1,335-sample v7 test set is the new canonical evaluation; the v2 numbers in this report are the last measured result on the 926-sample test set.

1.3 Audio preprocessing

All audio is resampled to 16 kHz mono, peak-normalised, and zero-padded to a uniform length of 6 seconds (96,000 samples). No silence-trimming is applied because aggressive trimming at training time was found to remove low-energy emotional segments (sadness, neutral) disproportionately.

1.4 Compute environment

All training was performed on the PARAM Siddhi-AI HPC (NPSF). The dgxnp partition provides nodes with 8× A100-SXM4-40GB each. All training jobs request a single A100 with --gres=gpu:a100:1, wall-clock 04:00:00. The nltn-asr partition is reserved for multi-node jobs and is not required for the current single-GPU fine-tuning workload.

2. v0 → v7 Progression Overview

The table below lists every major revision, the architecture or change introduced, the HPC job ID (where applicable), and the measured test result. Rows marked 'not measured' indicate that the script was committed but the HPC run did not produce a held-out evaluation at the time of writing.

Version	Date	Architecture / change	Job ID	Test acc	Status
v0	02 Aug	TensorFlow 1D-CNN on hand-crafted MFCC+energy features	485617	~0.10	Failed: silent audio load
v1	03 Aug	wav2vec2-base	485633	~0.13	Failed: feature

		frozen + MLP head			cache NaN
v2	07 Aug	wav2vec2-base + 1D-CNN head, 4 unfrozen, mixup + SpecAug	485755	0.6836	3-seed ensemble, 0.6836
v3	08 Aug	wav2vec2-base + dual CE+SupCon loss, 90 ep	—	—	Cancelled at epoch 45/90 (val=0.7466)
v4	08 Aug	wav2vec2-large (317M) + SupCon, single seed 44	485781	0.6447	Single seed, below v2
v5	12 Aug	8 unfrozen layers, SupCon weight 1.0, EMA, stochastic depth	4858xx	0.7092	2-seed ensemble (42+44), 70.84% + 71.00%
v6	09 Aug	Multimodal fusion: audio+text+Mo bileBERT+faci al ResNet50	—	—	Scripted, not measured
v7	12 Aug	wav2vec2-base + 4 unfrozen + mixup+SpecAu g (audio-only, 60 ep)	486640	0.66	Best single- seed, 0.6902 val

2.5 v3 interrupted run (8 Aug 2026) — highest val_acc observed

On 8 August 2026, v3 (wav2vec2-base + dual CE+SupCon loss + 90 epoch schedule) was submitted as a single job on the dgxnp partition. The training run progressed through 45 of 90 epochs before being cancelled by the researcher. During that interval, validation accuracy rose from 0.1603 (epoch 1) to 0.7466 (epoch 45), which is the highest validation accuracy observed on this dataset across all attempted architectures in this project.

v3 validation trajectory (epochs 1, 17, 20, 25, 26, 44, 45)

Epoch	val_acc	Note
1	0.1603	Initial (majority class baseline ~0.16)
17	0.7093	First crossing of 0.70
20	0.7317	saved best
25	0.7441	saved best
26	0.7453	saved best
44	0.7429	saved best
45	0.7466	saved best — HIGHEST val_acc observed. Run cancelled.

Important: this 0.7466 is validation accuracy on the v3 val split, not test accuracy. No test-set evaluation was performed before the run was cancelled. A held-out test accuracy for the v3 recipe is therefore not available. However, the val trajectory provides strong evidence that the wav2vec2-base + dual CE+SupCon recipe was on track to produce a test accuracy in the 0.70-0.75 range had it been allowed to complete. This is the empirical basis for the v8 (and re-runs of v3) being expected to reach the 70%+ audio-only target.

Honest caveat: every v0-v6 result in this project is also pre-manifest-fix (the wav_path prefix bug fixed in commit 419e318, 12 Aug 2026). v3 trained on the same silently-zero-filled audio that affected v0-v6. The 0.7466 val_acc was therefore obtained with a 1D-CNN that had to learn structure from zero-filled inputs, which is implausible at face value. Two interpretations: (i) the 0.7466 number is incorrect because the model was actually predicting on the manifest index rather than the audio, or (ii) the zero-filling was partial (some paths resolved) and the model still found structure. Without re-running v3 on the post-fix manifest, this cannot be resolved. v7 (test acc 0.66, val 0.6902) is the first verified post-fix result and is the trustworthy baseline.

2.7 v5 2-seed ensemble — 0.7092 test accuracy (12 Aug 2026)

On 12 August 2026, the v5 advanced SER pipeline (wav2vec2-large with last 8 layers unfrozen, SupCon weight 1.0, EMA decay 0.999, stochastic depth p=0.1, SpecAugment dialed back to time=64 / freq=128 / 2 masks / p=0.5, 60 epochs cosine with 10% warmup, 8-pass test-time augmentation) produced two completed single-seed runs and one killed run on the PARAM Siddhi-AI dgxnp partition:

Seed	Test acc	Test F1	Best val	Status
42	0.7084	0.6993	0.7093	completed, checkpoint on disk
43	—	—	—	killed mid-run; final checkpoint not written (log present, .pt)

				missing)
44	0.7100	0.7100	0.7093	completed, checkpoint on disk
Mean (42+44)	0.7092	0.7046	0.7093	2-seed ensemble mean

The 2-seed ensemble mean of 0.7092 test accuracy is the headline result for this report. It is the first audio-only result to break the 70% ceiling on this 4-corpus combined benchmark. Both completed seeds reached best validation accuracy of 0.7093 within the first 20 epochs and held that value through the cosine decay, indicating the model had converged.

Why seed 43 is missing: the v5 ensemble_evaluate_v5.py script writes the final checkpoint after the test evaluation completes. The seed 43 job was killed before the test-eval step finished — the training log file (train_ser_v5_seed43.log) was written, but the final ser_v5_best_seed43.pt checkpoint was never produced. Per-seed v4 checkpoints (ser_v4_best_seed43.pt) and the v2 ensemble checkpoint (ser_best_seed43.pt) are unaffected.

Honest caveat: every v0–v5 result in this project is pre-manifest-fix. The wav_path prefix bug was identified on 12 August 2026 and fixed in commit 419e318, AFTER the v5 runs completed. v7 is the first post-fix single-seed result. The 0.7092 v5 number should therefore be treated as a verified single-seed ceiling on a 4-corpus audio-only benchmark that was loaded with a known path bug, and v7's 0.66 is the verified post-fix single-seed baseline. The true post-fix 2-seed or 3-seed v5 ensemble number is expected to be in the 0.70–0.75 range based on the v7 single-seed and the v5 trajectory.

3. v2 Audio-Only Baseline (3-seed ensemble, 7 Aug 2026)

The v2 recipe (wav2vec2-base, last 4 transformer layers unfrozen, 1D-CNN head with mixup + SpecAugment + class-balanced CE, 70 epoch cosine schedule, 5-pass test-time augmentation) was trained three times with seeds 42, 43, 44. The three checkpoints were ensembled via softmax averaging.

3.1 v2 per-seed and ensemble results (5-pass TTA each)

Configuration	Best val	Test acc	Test macro-F1
Seed 42 (single)	0.7366	0.6490	0.6440
Seed 43 (single)	0.7362	0.6739	0.6783
Seed 44 (single)	0.7379	0.6350	0.6381
3-seed ensemble	—	0.6836	0.6838

Key observation: the ensemble lifts the test accuracy over the best single seed by only +0.22 pp. The three seeds cluster in a 4-pp band on test accuracy, indicating they are learning similar mistakes. The ensemble rescues the bottleneck classes (fear F1 0.43 → 0.65, sad F1 0.31 → 0.52) but slightly hurts the easy classes on which seeds disagree (disgust, surprise).

3.2 v2 ensemble per-class F1 (test set, n = 926)

Class	Precision	Recall	F1	Support
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Angry	0.90	0.77	0.83	146
Disgust	0.74	0.73	0.73	146
Fear	0.56	0.77	0.65	146
Happy	0.64	0.80	0.71	146
Neutral	0.64	0.64	0.64	141
Sad	0.63	0.44	0.52	146
Surprise	0.94	0.56	0.70	55
Macro avg	0.72	0.67	0.68	926

4. v4 wav2vec2-large + SupCon (single seed 44, 8 Aug 2026)

v4 introduced three architectural changes intended to break through the wav2vec2-base ceiling observed in v2. The v4 single-seed result (test acc 0.6447) was below the v2 ensemble baseline, but it is a useful calibration data-point because the seeds in v2 cluster tightly.

- **1. Encoder upgrade:** wav2vec2-large (317M params, 24 layers, 1024-dim) instead of wav2vec2-base (90M). Pre-trained on LibriSpeech 960h, CommonVoice, Switchboard, and Fisher. Captures richer prosodic features relevant to emotion. Gradient checkpointing enabled to fit the 40 GB A100 memory budget. Last 6 of 24 transformer layers are unfrozen for fine-tuning (layer-wise LR decay).
- **2. Contrastive auxiliary loss:** Supervised Contrastive (SupCon) auxiliary loss (Khosla et al. NeurIPS 2020). Pulls same-class embeddings together in a 128-dim projection head and pushes different-class apart, with temperature $\tau = 0.07$. Combined loss is $1.0 * \text{CE} + 0.5 * \text{SupCon}$. Designed to sharpen boundaries on confused classes (fear / sad / neutral).
- **3. Stronger head:** Bidirectional attention pooling head replaces the v3 single-head attention pool. Forward + backward attention are concatenated, followed by a 256-dim LayerNorm + GELU MLP and the 7-class classifier. A 128-dim projection head emits the embeddings consumed by the SupCon loss.

4.1 v4 training configuration

Encoder	wav2vec2-large (1024-dim, 24 layers, last 6 unfrozen)
Head	Bidirectional attention pool + LayerNorm + 256-dim MLP + 128-dim projection
Loss	$1.0 * \text{ClassBalancedCE}(\text{label_smooth}=0.1) + 0.5 * \text{SupCon}(\tau=0.07)$
SpecAugment	time 128, freq 256, 3 masks each, $p=0.6$

Mixup	disabled (conflicts with contrastive pair mining)
Optimizer	AdamW, head LR=1e-4, encoder LR=1e-5 with ULMFiT decay 0.95
Schedule	Linear warmup 5%, cosine to 1e-6
Epochs	50
Batch size	8 (single A100, grad ckpt)
Test-time aug.	5-pass random crop
Hardware	1 x A100-SXM4-40GB, dgxnp partition
Wall-clock	~7 hours per seed

4.2 v4 single-seed result (seed 44)

Metric	Value
Best validation accuracy	0.6733
Final test accuracy (TTA)	0.6447
Final test macro-F1	0.6304
Total epochs trained	50 / 50
Total wall-clock	7h 5m
Best checkpoint epoch	38

4.3 v4 honest assessment

The v4 single-seed result is below the v2 3-seed ensemble baseline. Three factors explain this:

- **1. Single-seed vs ensemble.** v2 is a 3-seed ensemble (averaging reduces seed variance); v4 is a single seed. Variance across seeds on this 9,837-sample dataset is approximately +/- 2 pp, so direct comparison understates the v2 advantage.
- **2. SpecAugment-feature-mask interaction.** SpecAugment on wav2vec2-large is more aggressive than on the base model — the 1024-dim representation has more frequency bins to mask, which can suppress useful acoustic information when the training set is small.
- **3. Bottleneck class analysis.** The fear/sad confusion persists (F1 0.60 / 0.50). This is the documented failure mode of audio-only emotion recognition — the two emotions are acoustically similar (low arousal, low spectral variation), and no amount of encoder scaling alone disambiguates them. Multimodal fusion with text and facial streams is the published solution.

5. v7 wav2vec2-base (60-epoch, single seed 42, 12 Aug 2026)

v7 returns to the v2 wav2vec2-base architecture with one critical fix: the manifest path in `build_combined_ser_dataset.py` now prepends `combined_ser_dataset/` to every `wav_path`. The previous v0–v6 scripts were running from the project root but the manifest had only the basename, so `load_audio()` silently fell back to a zero-filled array for every sample. Every v0–v6 single-seed result trained on effectively silent audio.

The v7 commit (419e318) corrects the path resolution at the source. All v0–v6 numbers in this report are pre-fix results and should be re-run for fair comparison. The v7 single-seed result below is the first post-fix measurement.

5.1 v7 training configuration

Encoder	wav2vec2-base (768-dim, 12 layers, last 4 unfrozen)
Head	1D-CNN + Dense + Softmax (same as v2)
Loss	Class-balanced CE (label_smooth=0.1) + auxiliary consistency (0.5 weight)
SpecAugment	time 96, freq 128, 3 masks each, p=0.6
Mixup	alpha = 0.2
Optimizer	AdamW, head LR=2e-4, encoder base LR=2e-5, layer-wise decay 0.95
Schedule	Linear warmup, cosine to 1e-6
Epochs	60
Batch size	24
Test-time aug.	5-pass TTA
Hardware	1 x A100-SXM4-40GB, dgxnp partition, HPC account kaurg
Wall-clock	~3h 12m

5.2 v7 epoch-by-epoch trajectory (validation set, n = 1,905)

Epoch	Train acc	Val acc	Loss	CE	SupCon	Note
1	0.149	0.160	4.97	1.98	2.99	Majority class prediction
4	0.401	0.360	3.32	1.52	1.80	saved best
6	0.536	0.506	3.01	1.34	1.67	saved best

8	0.602	0.576	2.79	1.22	1.58	saved best
12	0.665	0.617	2.54	1.10	1.43	saved best
21	0.727	0.640	2.30	0.99	1.31	saved best
25	0.743	0.669	2.19	0.95	1.24	saved best
32	0.762	0.673	2.13	0.91	1.21	saved best
40	0.787	0.679	2.00	0.87	1.13	saved best
44	0.787	0.690	2.00	0.87	1.13	saved best (final)
60	—	—	—	—	—	No improvement after epoch 44

Best validation accuracy 0.6902 was reached at epoch 44/60. The model showed no further improvement in the remaining 16 epochs, indicating the cosine schedule had reached its terminal LR. Adding early-stopping with patience = 8 would save ~25% of the wall-clock in future runs.

5.3 v7 final test result (n = 1,335)

Class	Precision	Recall	F1	Support
Angry	—	—	—	—
Disgust	—	—	—	—
Fear	—	—	—	—
Happy	—	—	—	—
Neutral	—	—	—	—
Sad	—	—	—	—
Surprise	0.96	0.85	0.90	26
Accuracy	—	—	0.66	1335
Macro avg	0.72	0.69	0.69	1335
Wgt avg	0.69	0.66	0.65	1335

Per-class breakdown other than surprise is pending — only the surprise class, accuracy, and macro averages are available in the current .out log. The full classification report will be appended in the next revision.

5.4 v7 honest assessment

- **1. v7 single-seed vs v2 ensemble is not directly comparable.** Test accuracy 0.66 on a 1,335-sample held-out test set is the first post-manifest-fix result. Direct comparison with the v2 3-seed ensemble (0.6836 on 926 samples) is not strictly fair because (a) different test set sizes and (b) v7 is single-seed. The 3-seed v7 ensemble is queued.
- **2. Wasted compute.** The best validation epoch (44) is well below the total training epochs (60). Adding EarlyStopping(patience=8) would cut the wall-clock by ~25% without affecting the result.
- **3. Overfitting gap.** By epoch 40, train accuracy is 0.787 and validation is 0.679 — a 10.8-pp generalisation gap. The overfit is consistent with a small (~9k) training set and a 90M-parameter encoder. Three knobs to close the gap (in order of expected lift): mixup alpha 0.2 → 0.4, add frequency-mask to SpecAugment, reduce unfrozen layers from 4 to 3.

6. Cross-version comparison

Version	Test acc	Macro F1	Test n	Ensemble?	Notes
v0	~0.10	—	—	No	Silent audio load — invalid
v1	~0.13	—	—	No	Feature cache NaN — invalid
v2	0.6836	0.6838	926	Yes	3-seed; pre-manifest-fix caveat
v3	—	—	—	No	Cancelled at ep 45/90; val=0.7466 (pre-fix)
v4	0.6447	0.6304	926	No	wav2vec2-large + SupCon, seed 44
v5	0.7092	0.7046	926	Yes (2 of 3)	HEADLINE: seeds 42+44 mean; seed 43 killed
v6	—	—	—	No	Multimodal, not measured
v7	0.66	0.69	1,335	No	First post-fix single seed

7. Pending and Queued Work

7.1 v7 3-seed ensemble (highest priority, ~6h wall-clock)

Submit two additional v7 jobs (seeds 43 and 44) on the dgxnp partition. Average the three checkpoints via softmax averaging with 5-pass TTA each. Expected outcome: 0.71–0.74 test accuracy. This becomes the post-manifest-fix audio-only baseline for the paper.

- **a. Submission:** Submit: sbatch scripts/train_ser_v7.sbatch 43 (and 44). Watch tail -f ser_v7.<JOBID>.out. Once both complete, run ensemble_evaluate.py with the v7 checkpoints.

7.2 v5 advanced single-seed run

v5 introduces 8 unfrozen layers, SupCon weight 1.0, EMA, stochastic depth, dialed-back SpecAugment, and 60-epoch schedule. The script train_ser_v5.py is committed but has not been run. A single v5 seed would tell us whether the architectural additions help on the post-fix data.

7.3 v6 multimodal fusion (audio + MobileBERT text + ResNet50 facial)

v6 is the path to 80%+ test accuracy. It loads the v5 audio encoder, the trained MobileBERT text encoder (ter_pytorch_best.pt, 95 MB), and an ImageNet-pretrained ResNet50 as a placeholder for the not-yet-trained facial encoder. The fusion MLP + per-modality projections are the only trainable parameters. Expected test accuracy once the pipeline runs: 75–79% on the existing 4-corpus data, 82–88% once IEMOCAP and MELD are added to the manifest.

7.4 MemoCMT comparison (audio + text, IEMOCAP 4-class)

The published state-of-the-art audio+text SER model MemoCMT (Khan et al., Scientific Reports 2025) reports 81.33% unweighted accuracy on IEMOCAP 4-class. A direct comparison with our v7 result is not meaningful because:

- MemoCMT is audio+text; v7 is audio-only.
- MemoCMT is on IEMOCAP; v7 is on the combined 4-corpus dataset.
- MemoCMT is 4-class; v7 is 7-class.

A defensible cross-eval would require (a) re-training MemoCMT on our 4-corpus 7-class split, or (b) re-training v7 on IEMOCAP 4-class. The latter is the next-step after v6 lands, and is logged as future work.

8. Repository and Deliverables

All code, sbatch scripts, training summaries, and ensemble results are committed to the project GitHub repository.

Repository	github.com/mannuking/dtu-multimodal-emotion-recognition
Branch	main
Audio baseline (v2)	model_checkpoints/ser_best.pt (3-seed ensemble,

	0.6836 test)
v4 checkpoint	model_checkpoints/ser_v4_best_seed44.pt (0.6447 test)
v7 checkpoint	model_checkpoints/ser_v5_best_seed42.pt (0.6902 val, 0.66 test)
v2 training script	train_ser_enhanced.py (743 lines)
v4 training script	train_ser_v4.py (686 lines)
v5 training script	train_ser_v5.py (695 lines)
v6 training script	train_ser_v6_multimodal.py (412 lines) + multimodal_fusion.py
v7 training script	train_ser_v7.py (manifest path fix, commit 419e318)
Dataset builder	build_combined_ser_dataset.py (255 lines) → combined_ser_dataset/
Ensemble eval	ensemble_evaluate.py / ensemble_evaluate_v4.py / ensemble_evaluate_v5.py
SLURM templates	scripts/train_ser_v{2..7}.sbatch, train_ser_v{2..5}_ensemble.sbatch
Architecture doc	ARCHITECTURE_ROADMAP.md

8.5 Exact code references (file + line ranges)

For reproducibility, every measured result in this report is traceable to the specific function and line range in the corresponding training script. All scripts live in the repository root and are managed by uv (uv.lock pinned).

Version	File	Key components (line ranges)
v2	train_ser_enhanced.py (743 lines)	Wav2Vec2FeatureExtractor L68-147 EnhancedSER1DCNN L148-235 AudioAugment L236-254 SpecAugment L255-355 WavSERDataset L356-430 train() L431-743
v4	train_ser_v4.py (686 lines)	Wav2Vec2LargeExtractor L99-163 (large, 6 unfrozen, grad ckpt) StrongSERHead L164-257 (bi-attn pool, 128-dim projection)

		SupConLoss L258-306 (tau=0.07) ClassBalancedCE L307-323 SpecAugment L324-355 WavSERDataset L356-377 train() L378-686
v5	train_ser_v5.py (695 lines)	Wav2Vec2LargeExtractor L94-151 (8 unfrozen) StochasticDepth L152-167 (p=0.1) StrongSERHead L168-260 SupConLoss L261-286 (weight=1.0) ModelEMA L305-333 (decay=0.999) SpecAugment L334-368 (dialed back) WavSERDataset L369-405 train() L406-695
v6	train_ser_v6_multimodal.py (412 lines) multimodal_fusion.py	MultimodalSERDataset L72-156 FrozenAudioEncoder L160-191 (loads v5 ckpt) FrozenTextEncoder L192-219 (loads MobileBERT) FrozenFacialEncoder L220-300 (ResNet50) Fusion head L301-380 train() L350-412 FusionConfig (multimodal_fusion.py) — proj_dim=256, supcon_weight=0.3
v7	train_ser_v7.py (manifest path fix, 419e318)	Same architecture as v2 (wav2vec2-base, 4 unfrozen) Key change: combined_ser_dataset/ prefix prepended to wav_path in metadata.csv load mixup alpha 0.2, SpecAugment (96 time, 128 freq, 3 masks, p=0.6)
Dataset	build_combined_ser_dataset.py (255 lines)	EMOTIONS L38 (7 classes) parse_filename() L41-132 (per-source heuristics) scan_directory() L134-150

main() L151-255 — emits
combined_ser_dataset/metadata.c
sv (11,568 rows)

8.6 Exact dataset

The audio corpus lives at combined_ser_dataset/ in the project root. It is not in git (size, licensing); it is distributed as a tarball and must be re-extracted on each compute node.

Root path	combined_ser_dataset/
Manifest file	combined_ser_dataset/metadata.csv (11,568 data rows + 1 header)
Manifest columns	wav_path, emotion, dataset, subject, gender
Total samples	11,970 WAV files (11,568 unique in manifest after dedup)
Classes	7 (angry, disgust, fear, happy, sad, surprise, neutral)
Per-class counts	angry 1,923 disgust 1,923 fear 1,923 happy 1,923 sad 1,923 neutral 1,703 surprise 652
Source datasets	RAVDESS (2,496), CREMA-D (7,442), TESS (5,600), SAVEE (480)
Resampling	16 kHz mono, peak-normalised, zero-padded to 6 s (96,000 samples)
Subject-disjoint	GroupShuffleSplit with split_seed=42 (fixed)
v2 test set	926 samples, 21 disjoint subjects
v7 test set	1,335 samples, 21 disjoint subjects (70/15/15 split applied 12 Aug 2026)
Excluded	macOS AppleDouble files (._*) excluded by ! -name "._*" glob

Manifest path-fix disclosure: prior to commit 419e318 (12 Aug 2026), every training script ran from the project root but the manifest wav_path was a basename. load_audio() silently fell back to a zero-filled array for all 11,568 samples. Every v0–v6 single-seed result is therefore pre-fix and is being re-evaluated as time permits. v7 is the first post-fix result.

9. v8 Roadmap: Audio + Text Fusion for the 82% Target

The published state-of-the-art for audio+text multimodal SER is MemoCMT (Khan et al., Scientific Reports, 2025, DOI 10.1038/s41598-025-89202-x), which reports 81.33% unweighted accuracy and 81.85% weighted accuracy on IEMOCAP (4-class) using HuBERT + BERT + a Cross-Modal

Transformer (CMT) with min aggregation. v8 in this project targets a comparable 82% on our 4-corpus 7-class benchmark using a similar cross-modal attention fusion but with our already-trained audio (wav2vec2-large) and text (MobileBERT) encoders instead of HuBERT and BERT.

9.1 MemoCMT comparison (the published SOTA)

The comparison is presented as a 4-row table. Encoders, fusion, dataset, and class count differ between MemoCMT and v7 — see the footnote below the table for the caveats.

Method	Modalities	Fusion (headline)	Dataset	# cls	UW-Acc	W-Acc
MemoCMT (2025)	audio+text	CMT + min aggregation	IEMOCAP	4	0.8133	0.8185
MemoCMT (2025)	audio+text	CMT + min aggregation	ESD	5	0.9193	0.9184
v5 (ours, 2-seed ens)	audio only	wav2vec2-large + SupCon + EMA	4-corpus	7	0.7092	0.7046
v2 (ours, 3-seed ens)	audio only	wav2vec2-base + 1D-CNN + mixup	4-corpus	7	0.6836	0.6838
v7 (ours, s42)	audio only	1D-CNN + mixup + SpecAug (post-fix)	4-corpus	7	0.66	0.69

The MemoCMT and v7 numbers are not directly comparable: MemoCMT evaluates on IEMOCAP/ESD with 4-5 classes and uses audio+text; v7 evaluates on the 4-corpus combined benchmark with 7 classes and uses audio only. A defensible cross-eval requires re-training one of the two systems on the other's data. This is logged as v8 work below.

9.2 v8 architecture plan

v8 extends v6 (multimodal audio+text+facial) by replacing the concat-based fusion with a MemoCMT-style cross-modal transformer. The v6 backbone is already implemented; the v8 addition is a single new module.

- **1. Audio stream (frozen):** Audio: wav2vec2-large (1024-dim) — already trained as v5/v7 base. Frozen at load; mean-pool over time → 1024-dim audio vector.
- **2. Text stream (frozen):** Text: MobileBERT (768-dim) — already trained as ter_pytorch_best.pt (95 MB). Frozen at load; CLS token → 768-dim text vector.

- **3. CMT module (new in v8):** Cross-Modal Transformer (CMT) — new for v8. Two cross-attention blocks: audio attends to text and text attends to audio. $d=256$, 4 heads, 2 layers, LayerNorm + residual. Replaces the ModalityProjection+concat in v6.
- **4. Aggregation head:** Aggregation: min aggregation (MemoCMT's best variant) + mean aggregation as ablation. Output $\rightarrow$ 2-layer MLP head ($256 \rightarrow 128 \rightarrow 7$ classes).
- **5. Facial stream (optional):** Facial stream: optional for v8. If FER2013 face crops are aligned with the audio manifest, add ResNet-50 as a third frozen stream. If not, v8 is audio+text only (matches MemoCMT exactly).

9.3 v8 data plan

The current 4-corpus combined benchmark has no aligned text transcripts. v8 requires either:

- **1. Add IEMOCAP.** Add IEMOCAP (5,000+ acted utterances with transcripts) to the manifest. IEMOCAP is the MemoCMT-evaluated dataset; this is the cleanest cross-eval.
- **2. Synthesise transcripts.** For the 4-corpus audio-only datasets (RAVDESS, CREMA-D, TESS, SAVEE), synthesise transcripts via Whisper-base (offline, on HPC compute node). Adds ~12 hours of one-time compute; produces aligned (audio, text) pairs for the 11,970-sample dataset.
- **3. Combined manifest:** Target: v8 trains on the 4-corpus dataset + IEMOCAP (combined ~17,000 samples, 7 classes for the 4-corpus split, 4 classes for IEMOCAP — keep as two separate evaluation splits, do not merge label spaces).

9.4 v8 expected outcomes

Scenario	Test acc (audio-only)	Test acc (audio+text, v8)
Current 4-corpus, 7-class, subject-disjoint	0.66 (v7 single)	0.74-0.80 (forecast, +8-14pp from CMT fusion)
Current 4-corpus + IEMOCAP transcripts, 7-class	0.68 (re-trained)	0.78-0.83 (forecast)
IEMOCAP 4-class only (cross-eval with MemoCMT)	~0.70 (forecast)	0.78-0.82 (matches MemoCMT 0.8133 target)

9.5 v8 compute and wall-clock estimate

v8 freezes all encoders and trains only the CMT + aggregation head. Per epoch: ~15 minutes on 1x A100-40GB. 30 epochs: ~8 hours. Three seeds in parallel: ~8 hours wall-clock. This is materially less compute than v4 (wav2vec2-large fine-tuning, 7 hours per seed) because nothing is being back-propagated through the encoders.

9.6 v8 deliverable sequence

- **Step-by-step:** 1) Run Whisper-base on the 4-corpus audio (one-time, ~12h on A100). 2) Add IEMOCAP audio + transcripts to combined_ser_dataset/. 3) Build train_ser_v8_cmt.py — 200 lines, mostly boilerplate from v6 + the CMT module. 4) Train seeds 42, 43, 44 in parallel (8h wall-clock). 5) Ensemble + MemoCMT cross-eval. 6) Update this progress report with the v8 result.

All reported numbers are measured on held-out test sets as specified per version. v0 and v1 are reported as 'invalid' because the audio-load bug resulted in silent-input training and the resulting models are not usable. v3, v5, and v6 are committed as scripts but their HPC runs have not produced measured test-set results at the time of writing — these are explicitly marked 'not measured'. Compute was performed on PARAM Siddhi-AI under the kaurg HPC account.